



From Costs to Gains: How Cost of Sales Enhances Firm Value in Listed Agricultural Companies



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Abstract: This study examined how cost of sales influences the firm value of listed agricultural companies in Nigeria. An ex-post facto research design was adopted to analyze audited historical financial data collected from five listed Nigerian agricultural companies, including Ellah Lakes PLC, FTN Cocoa Processors PLC, Livestock Feeds PLC, Okomu Oil Palm PLC, and Presco PLC, which were selected by census sampling. The secondary data obtained from the annual reports of the firms under investigation was from the period of 2015 to 2024. Hypotheses were tested using panel estimated generalized least squares. The findings revealed that cost of sales had a significantly positive effect on firm value ($\beta = 8.801653$, $p = 0.0000$), indicating that effective management of production and operational costs enhanced financial returns. Therefore, the management of listed agricultural companies was advised to strengthen structured cost management practices that focused on efficient procurement of raw materials, improved inventory control, and optimized production processes, so that spending on cost of sales continued to support the growth of revenue and translate into higher firm value rather than generating unnecessary operational waste.

Keywords: Cost of sales; Firm value; Agricultural firms

1. Introduction

Agricultural companies play an essential role in the economic development of many countries, to provide food security, employment opportunities, and raw materials for other industries. In Nigeria, the agricultural sector remains one of the largest contributors to gross domestic product and has significant potential for further growth, especially through value addition and efficient resource management. Listed agricultural firms are increasingly operating in a highly competitive environment where profitability and sustainability depend on effective management of operational costs, particularly cost of sales (Leting, 2024). Cost of sales, also known as cost of goods sold, represents the direct expenses incurred in producing or acquiring the goods that a company sells during a specific period (Giwa & Dibua, 2025). These costs include expenses such as raw materials, direct labor, and production overheads. Managing these costs efficiently has become a pressing concern for agricultural firms as they face challenges such as fluctuating input prices, supply chain disruptions, and changing consumer demands. This study focused on understanding how cost of sales could influence the financial performance and overall firm value of listed agricultural companies. Agricultural companies that fail to control the risk associated with cost of sales would reduce profitability, limit investment capacity, and lower shareholder returns; thus the study of this topic was considered both timely and essential.

Firm value is a measure of the overall worth of a company in terms of its ability to generate wealth for shareholders (Ukoh et al., 2024). It reflects the combined effects of financial performance, market perception, and strategic decision making. In today's business environment, firm value has become a key indicator for investors, stakeholders, and managers because it captures both tangible and intangible aspects of business success. Shareholders and potential investors often look at firm value to assess the profitability, potential of growth, and risk profile of companies before committing resources (Oreshile et al., 2025). For agricultural companies, firm

value is not only a measure of financial success but also a reflection of operational efficiency, sustainability practices, and market competitiveness. Cost of sales is closely related to firm value because it directly affects profit margins, cash flow, and investment capacity (Giwa & Dibua, 2025). When a company manages its cost of sales effectively, it reduces waste, optimizes utilization of resources, and improves production efficiency, all of which contribute to higher profitability. This, in turn, could enhance investor confidence, attract new capital, and increase stock prices for listed companies. Conversely, high or poorly managed costs of sales erode profits, create financial instability, and reduce the attractiveness of a company to investors. In today's market, where agricultural firms face rising input costs, volatile commodity prices, and increasing pressure to adopt sustainable practices, monitoring and controlling cost of sales is critical. Understanding the link between cost of sales and firm value allows managers to implement strategies that maximize returns while maintaining competitive pricing and operational efficiency. This connection highlights the importance of cost management as a strategic tool for enhancing financial performance and building long-term value in listed agricultural firms.

Cost of sales enhances firm value by directly influencing profitability, operational efficiency, and investors' perception. When agricultural companies reduce unnecessary expenses in raw materials, labor, and production overheads, they increase net profit margins, which have a positive effect on retained earnings and shareholder returns. Higher profitability attracts investors, raises stock prices, and strengthens the company's market position (Nalurita, 2017). Efficient management of cost of sales allows firms to allocate resources toward research and development, expansion projects, and adoption of sustainable practices; all of which contribute to long-term value creation (Giwa & Dibua, 2025). For example, by sourcing raw materials strategically and investing in modern production techniques, agricultural firms could reduce production costs, improve product quality, and increase output, which enhances revenue and overall competitiveness in the market. Moreover, effective management of cost of sales helps firms navigate economic challenges such as inflation, disruptions in the supply chain, and fluctuating demand (King et al., 2025). Companies are better equipped to sustain operations during periods of financial uncertainty by maintaining optimal cost structures, which reassures investors and other stakeholders of the firm's stability. Beyond financial metrics, cost of sales affects firm value through its impact on operational decision making and strategic planning. Companies that monitor and analyze their cost of sales could identify inefficiencies, streamline processes, and implement cost-saving measures without compromising product quality. This continuous improvement not only supports profitability but also builds confidence among investors and partners, thus contributing to higher valuation in the capital market (Question & Atagboro, 2025).

However, many agricultural companies in practice appear to face substantial challenges in controlling their cost of sales (Leting, 2024). Fluctuating prices of raw materials, inefficiencies in labor management, high production overheads, and inadequate coordination of the supply chain often leads to escalating production costs. Some firms struggle to implement modern production techniques or cost monitoring systems, resulting in expenses that reduce net profit margins. These challenges prevent agricultural companies from fully translating sales revenue into firm value. While some companies generate substantial revenue, their profitability remains low, most probably due to the disproportionate increase in costs relative to sales (Ismanov et al., 2024). Low profitability limits the ability of firms to reinvest in production, research, and innovation, hence restricting their potential of growth. Shareholders experience lower returns, and the market may perceive the companies as less attractive investments, which could depress stock prices and overall firm value (Giwa & Dibua, 2025). Additionally, inadequate cost control could lead to financial instability, rendering it difficult for companies to withstand economic fluctuations or unexpected disruptions in the agricultural sector. Over time, persistent inefficiencies in managing cost of sales could erode competitiveness, reduce investor confidence, and undermine the long-term sustainability of listed agricultural firms.

Despite the growing body of research on the relationship between cost management and firm performance in Nigeria, a clear gap exists regarding the specific effect of cost of sales on the firm value of listed agricultural companies. Previous studies have largely focused on firms of manufacturing and consumer goods, with Adesina & Tihamiyu (2025); Akinleye & Fakorede (2025); Awotomilusi et al. (2022); Esangbedo & Adeyemi (2023); Giwa & Dibua (2025); King et al. (2025); Mamidu & Akinola (2019); Nkpodot & Emenyi (2023); Question & Atagboro (2025) examining various aspects of cost management, profitability, and shareholder returns in these sectors. Moguluwa et al. (2021) addressed cost and competitiveness in agricultural products but focused on farmers' export decisions rather than financial performance at firm level or value creation. While these studies provide useful hints on how cost components such as material, labor, overhead, and administrative expenses influence profitability and efficiency, they do not explicitly link cost of sales to firm value in the agricultural sector, which has distinct operational, market, and production characteristics compared to manufacturing firms. Most prior research relied on different financial metrics such as return on assets (ROA), return on equity, or net profit margins without fully integrating market-based measures of firm value. They also did not account for the unique dynamics of listed agricultural companies over a sufficiently long period. There is limited application of panel data techniques with corrections for heteroskedasticity and cross-sectional dependence in the context of agricultural firms, despite evidence from manufacturing firms that such methods yield more robust results. Therefore, there is a need to investigate how cost of sales, as a critical operational resource, affects firm value specifically in listed agricultural

companies in Nigeria, hence addressing both profitability and market-based indicators while controlling firm size (FSZ) and other contextual factors.

2. Literature Review

2.1 Synthesis of Existing Empirical Studies

Several studies have highlighted the significant role played by cost of sales and related cost management practices in shaping financial performance and shareholder value in Nigerian companies. Giwa & Dibua (2025) found that cost of sales positively and significantly affects shareholder returns, thus emphasizing the importance of strategic investments in the sourcing of raw materials, inventory control, and production efficiency. Similarly, King et al. (2025) reported a positive relationship between cost of sales and shareholders' wealth maximization among listed manufacturing firms, to support the notion that careful cost management could directly influence investors' returns. Question & Atagboro (2025) further demonstrated that while employee-related expenses had minimal effect on net profit margins, overall costs including cost of sales remained key determinants of financial performance. These studies collectively emphasized the centrality of cost of sales as a driver of profitability and shareholder value in Nigerian firms.

Beyond direct cost management, research has also explored the specific impact of material and overhead costs on firm profitability. Akinleye & Fakorede (2025) discovered that increases in overhead and material costs were associated with higher returns on assets, hence suggesting that investments in production inputs, when managed effectively, could enhance profitability. Adesina & Tiamiyu (2025) corroborated this by showing that administrative, marketing, distribution, and excessive production costs could erode profits, thus reducing unnecessary expenditures and implementing cost-efficient technologies could improve performance. Nkpodot & Emenyi (2023) similarly emphasized that efficiency in the management of raw materials and labor positively affected financial performance, in order to highlight the practical importance of operational cost controls in maximizing firm value. These findings suggested that not all cost increases were detrimental; rather, strategic allocation and monitoring of costs could contribute to higher profitability and enhanced market perception.

In the agricultural context, the influence of costs extends beyond profitability to competitiveness and strategic decision making. Moguluwa et al. (2021) demonstrated that cost and competitiveness, which accounted for a substantial portion of variance in export behavior, significantly shaped the export decisions of Nigerian farmers. Firms that manage costs effectively not only improve internal financial metrics but also strengthen their position in broader markets. Esangbedo & Adeyemi (2023) added that sales and distribution costs could have varying effects on revenue and gross margin, with high efficiency in these areas supporting revenue generation, though not always directly influencing ROA or return on equity. Such findings highlighted the nuanced nature of cost management, where both production and operational costs should be optimized to sustain long-term growth and firm value.

While most studies reported positive effects of cost management on firm performance, some evidence pointed to potential challenges when costs were poorly managed. Awotomilusi et al. (2022) found that direct cost structures negatively affected financial performance, thus emphasizing the risks of inefficient cost allocation. Mamidu & Akinola (2019) further reinforced the value of systematic cost management by demonstrating that effective oversight of material, labor, and overhead expenses enhanced operating profit and shareholder returns. Taken together, these empirical findings indicated that managing the cost of sales strategically was a critical pathway for enhancing firm value. Companies that balance cost control with investment in efficiency and competitiveness are better positioned to achieve sustainable profitability, strengthen shareholder confidence, and improve overall market valuation.

2.2 Theoretical Framework and Development of Research Hypothesis

The study was anchored on Resource-Based View (RBV) of firms, first formally introduced by Wernerfelt in 1984 (Lockett et al., 2009); the view was built upon earlier strategic management ideas that emphasized the importance of internal resources in creating competitive advantage (Raduan et al., 2009). The theory gained wider recognition in 1991 through the work of Barney, who provided a clear framework for understanding how resources and capabilities of a firm contributed to sustained performance. RBV focuses on the unique assets, skills, and capabilities that a firm possesses, arguing that these internal resources, if valuable, rare, inimitable, and non-substitutable, could provide the foundation for success in the long term. The theory marks a shift from examining external market conditions to emphasizing the strategic importance of what a firm owns and how it manages its resources to achieve superior performance.

The main postulations of the RBV centered on the idea that not all resources contributed equally to competitive advantage (Halawi et al., 2005). According to RBV, resources that are valuable allow a firm to implement strategies for improving efficiency and effectiveness. Rare resources provide a uniqueness that competitors cannot

easily replicate, hence creating differentiation in the market (Frances & Nworie, 2025). Resources that are difficult to imitate give the firm protection against competitive pressures, and non-substitutable resources ensure that the advantage cannot be replaced by alternative solutions. RBV also emphasizes the importance of organizational capability in combining and utilizing resources effectively, arguing that even valuable resources may fail to enhance performance if not managed properly. This framework highlights the role of strategic resource management as a key determinant of profitability, competitiveness, and long-term value creation.

The RBV is particularly relevant to this study because cost of sales represents one of the critical resources for agricultural companies that manage to create value. Under the RBV, capabilities of cost management can be conceptualized as strategic organizational routines that enable firms to coordinate procurement, production, and inventory processes more efficiently than competitors. These capabilities are valuable because they directly enhance margin generation and stability of cash flow, thereby strengthening the firm's capacity to create and sustain superior market value. They may also be rare and difficult to imitate when embedded in firm-specific systems, managerial expertise, supplier relationships, and internally developed operational technologies that competitors cannot rival. When such cost management processes are deeply integrated into the structure and culture of a firm, they become non-substitutable strategic assets, hence positioning cost efficiency not merely as an accounting outcome but as a Valuable, Rare, Inimitable, Non-substitutable-based source of sustained value creation. In line with the above theory, we hypothesized:

H1: Cost of sales will have a positive effect on the firm value of listed agricultural companies.

3. Methodology

This study adopted an ex-post facto research design to investigate how cost of sales influenced the firm value of listed agricultural companies in Nigeria. The ex-post facto design was considered appropriate because it involved analyzing historical data without manipulating any variables (Ikwor et al., 2025; Nworie et al., 2022; Nworie et al., 2025). This approach allowed the study to examine the relationship between cost of sales and firm value over a defined period, so as to capture the effects of cost management decisions made by the companies within the study period. The design facilitated a detailed assessment of financial records, to enable the identification of patterns and interrelationships that contribute to firm value in the agricultural sector.

The population of the study comprised all Nigerian Exchange Group listed agricultural companies that met the criteria for inclusion. Specifically, the companies included in the study were Ellah Lakes PLC, FTN Cocoa Processors PLC, Livestock Feeds PLC, Okomu Oil Palm PLC, and Presco PLC. These firms were chosen because they represented the primary players in the listed agricultural sector in Nigeria and had complete financial records for the period under review. Given the small number of eligible firms, census sampling was employed to ensure that all relevant companies were included in the study. By adopting this approach, the study avoided the potential bias that could arise from selecting only a subset of companies and ensured comprehensive coverage of the population.

The sample size therefore consisted of all five listed agricultural companies identified in the population. The study period spanned from 2015 to 2024, providing ten years of data for each firm. This time frame was selected to capture sufficient historical data to observe trends in cost of sales and firm value, as well as accounting for variations in market conditions, production costs, and financial performance over time. The use of longitudinal data allowed a more robust analysis of how changes in cost of sales affected firm value, while controlling other factors that might influence financial outcomes.

The method of data collection was secondary in nature, with financial information extracted from the annual reports of the selected firms for the period 2015 to 2024. The annual reports provided detailed data on cost of sales, total assets, revenue, net income, and other financial indicators necessary to measure firm value. To mitigate the risk of limited model specification and omitted variable bias, the study incorporated FSZ (measured as the natural logarithm of total assets), firm leverage (FLEV) (total liabilities divided by total assets), firm liquidity (FLIQ) (current assets divided by current liabilities), and firm profitability (ROA) as control variables. Including these factors helped ensure that the estimated effect of cost of sales on firm value was not confounded by other firm-specific characteristics that could also influence valuation. Reliability and consistency could be guaranteed despite the use of secondary data, as the information was audited and verified by the respective firms, thus providing an efficient means of collecting large amounts of historical financial data.

To examine the relationship between cost of sales and firm value, the study employed panel estimated generalized least squares as the primary method of data analysis. This technique was chosen because it could effectively handle panel data, which consisted of multiple observations across both time and entities, while addressing statistical issues such as heteroskedasticity and cross-sectional dependence. Heteroskedasticity, which arose when the variance of the error terms was not constant across observations, was corrected using White cross-section standard errors and covariance, to ensure that the estimates remained unbiased and efficient. Cross-sectional dependence, which occurred when error terms across different companies were correlated, was addressed using cross-section weights, to enhance the accuracy and reliability of the regression results.

The study specified the following regression model to test the hypotheses:

$$FV_{it} = \beta_0 + \beta_1 \text{COS}_{it} + \beta_2 \text{FSZ}_{it} + \beta_3 \text{FLEV}_{it} + \beta_4 \text{FLIQ}_{it} + \beta_5 \text{FPROF}_{it} + \varepsilon_{it} \quad (1)$$

where,

FV_{it} represents the firm value of company i at time t , measured using market capitalization.

COS_{it} represents the cost of sales for company i at time t , serving as the independent variable.

FSZ_{it} represents FSZ measured as the natural logarithm of total assets of company i at time t .

FLEV_{it} represents FLEV measured as total liabilities divided by total assets of company i at time t .

FLIQ_{it} represents FLIQ measured as current assets divided by current liabilities of company i at time t .

FPROF_{it} represents firm profitability measured as ROA of company i at time t .

β_0 is the intercept.

β_1 – β_5 are the coefficients of the explanatory variables.

ε_{it} is the error term.

The use of panel data regression allowed the study to capture both cross-sectional and temporal variations in the data, thus providing a more nuanced understanding of how cost of sales influenced firm value over time and across different companies. By including total assets as a control variable, the analysis accounted for FSZ, which is known to influence profitability, market perception, and overall valuation. This ensured that the effect of cost of sales on firm value was measured independent of variations in company size.

4. Data Analysis

4.1 Descriptive Analysis and Model Diagnostics

The descriptive statistics for market value in Table 1 show that the average market value of the listed agricultural companies was ₦56,382,750,000. This indicates that the market capitalization of most firms was around this level, in billions of naira, during the study period. The maximum market value was ₦585,000,000,000, while the minimum was ₦440,000,000, thus highlighting a very wide range of market values among the firms. The standard deviation of ₦110,908,689,000 reflects considerable dispersion from the mean. The skewness of 3.177 suggests a positively skewed distribution, meaning most firms had market values below the mean, while a few very high values pulled the average upward. The kurtosis of 13.723 indicates a leptokurtic distribution with heavy tails, showing that extreme values occurred more frequently than in a normal distribution. The Jarque-Bera statistic of 323.6912 with a probability of 0.000 confirms non-normality. Despite this, the Central Limit Theorem supports the reliability of the mean for analysis, as the sample of 50 observations allows approximation to normality.

Table 1. Descriptive statistics

	Market Value (₦'000)	Cost of Sales (₦'000)	FSZ	FLEV	FLIQ	ROA
Mean	56382750	7721702	7.173493	0.602275	1.577954	0.004214
Median	7738181	4579030	6.963047	0.576297	1.172195	0.023531
Maximum	585000000	49675266	8.518831	1.299719	11.54210	0.341408
Minimum	440000	0.000000	6.062894	0.050026	0.000418	-0.803813
Std. dev.	110908689	10330048	0.634664	0.270880	1.870983	0.196621
Skewness	3.177059	2.122333	0.156993	0.459226	3.732823	-1.385512
Kurtosis	13.72366	7.825315	2.028903	3.280501	18.96006	7.237857
Jarque-Bera	323.6912	86.04345	2.170035	1.921321	646.7902	53.41249
Probability	0.000000	0.000000	0.337896	0.382640	0.000000	0.000000
Observations	50	50	50	50	50	50

Note: FSZ = Firm size; FLEV = Firm leverage; FLIQ = Firm liquidity; ROA = Return on assets.

For cost of sales, Table 1 reports a mean of ₦7,721,702,000, indicating that, on average, firms spent this amount in billions of naira on producing and selling goods. The maximum cost of sales was ₦49,675,266,000, while the minimum was ₦0, demonstrating large differences across companies. The standard deviation of ₦10,330,048,000 further shows substantial variation. The skewness of 2.122 points to a positively skewed distribution, suggesting that most firms had cost of sales below the mean, with a few very high values increasing the average. The kurtosis of 7.825 indicates a leptokurtic distribution, showing that extreme values were more frequent than in a normal distribution. The Jarque-Bera statistic of 86.04345 with a probability of 0.000 confirms non-normality. Nevertheless, with 50 observations, the Central Limit Theorem ensures that the mean could still be used reliably in parametric analysis.

FLEV recorded a mean of 0.602275, implying that on average about 60% of total assets were financed through liabilities. The maximum value of 1.299719 suggests that in some instances liabilities exceeded total assets, while

the minimum of 0.050026 indicates very low debt usage for certain firm-years. The standard deviation of 0.27088 shows moderate variability in the financing structure. The skewness of 0.459226 points to a slight right skew, and the kurtosis of 3.280501 is close to 3, indicating an approximately mesokurtic distribution. The Jarque-Bera statistic of 1.921321 with a probability of 0.38264 confirms normality, and with 50 observations, the assumptions underpinning parametric panel estimation are further strengthened under the Central Limit Theorem.

FLIQ had a mean of 1.577954, indicating that on average current assets exceeded current liabilities. This suggests satisfactory short-term solvency among the firms. However, the maximum value of 11.5421 and the minimum of 0.000418 show extreme variation in liquidity positions, supported by a relatively high standard deviation of 1.870983. The skewness coefficient of 3.732823 reveals a highly right-skewed distribution, meaning a few observations with very high liquidity ratios influenced the mean, while the kurtosis of 18.96006 indicates extreme leptokurtosis with significant outliers. The Jarque-Bera statistic of 646.7902 and probability of 0.000000 confirm non-normality; nonetheless, the Central Limit Theorem justifies the use of inferential statistics given the adequate sample size.

Firm profitability (ROA) showed a mean of 0.004214, implying that on average the firms generated a modest return of about 0.4% on total assets during the period. The maximum value of 0.341408 and minimum of -0.803813 indicate substantial fluctuations in performance, including periods of significant losses. The standard deviation of 0.196621 reflects moderate dispersion relative to the mean. The negative skewness of -1.385512 suggests a left-skewed distribution, meaning more extremely negative profitability values occurred, while the kurtosis of 7.237857 indicates a leptokurtic distribution with heavy tails. The Jarque-Bera statistic of 53.41249 with a probability of 0.000000 shows that profitability was not normally distributed; however, with 50 observations, the Central Limit Theorem supports the robustness of the regression estimates despite departures from normality in the underlying data.

Table 2 presents the multicollinearity test using Variance Inflation Factors (VIF), which assessed the degree of linear association among the explanatory variables to determine whether multicollinearity might distort the regression estimates. The essence of this test was to ensure that the independent variables were not excessively correlated, as high multicollinearity could inflate standard errors and weaken the statistical significance of coefficients. As shown in the table, the centered VIF values for cost of sales (1.733493), FSZ (1.879687), FLEV (1.352583), FLIQ (1.038059), and firm profitability (1.721625) are all well below the conventional threshold of 10, and even below the stricter benchmark of 5, indicating the absence of harmful multicollinearity. This suggests that each explanatory variable contributed unique information to the model and that the estimated coefficients could be interpreted reliably.

Table 2. Multicollinearity test

Variable	Centered VIF
Cost of sales	1.733493
FSZ	1.879687
FLEV	1.352583
FLIQ	1.038059
ROA	1.721625
C	N/A

Note: VIF = Variance inflation factors; C = Constant term; FSZ = Firm size; FLEV = Firm leverage; FLIQ = Firm liquidity; ROA = Return on assets; Sample: 150.

Table 3. Cross-section dependence test

Test	Statistic	df	Prob.
Breusch-Pagan LM	48.43619	10	0.0000
Pesaran scaled LM	8.594593		0.0000

Note: LM = Lagrange multiplier; Null hypothesis: No cross-section dependence (correlation) in residuals; Equation: Untitled; Periods included: 10; Cross-sections included: 5; Total panel observations: 50; Non-zero cross-section means detected in data; Cross-section means were removed during computation of correlations.

Table 3 reports the Cross-section dependence test using the Breusch-Pagan lagrange multiplier (LM) statistic, which examined whether residuals across cross-sectional units (firms) were correlated. The essence of this test in panel data analysis was to verify whether shocks affecting one firm were correlated with shocks affecting others, since cross-sectional dependence could lead to biased standard errors and misleading inference if ignored. The null hypothesis stated that there was no cross-section dependence found in the residuals; however, the reported LM statistic of 48.43619 with a probability value of 0.0000 led to rejection of the null hypothesis at the 5% level. This indicates the presence of significant cross-sectional dependence among the sampled firms, meaning that common factors might simultaneously influence them. To address this issue, the model was corrected using cross-section Seemingly Unrelated Regression (SUR), thereby improving the efficiency and robustness of the estimates.

Table 4 shows the heteroskedasticity likelihood ratio test based on the Breusch-Pagan-Godfrey procedure, which evaluated whether the variance of the error terms was constant across observations. The essence of this test was to confirm the homoskedasticity assumption of classical regression, as heteroskedasticity could result in inefficient estimates and biased standard errors, ultimately affecting the testing of the hypothesis. The reported *F*-statistic of 11.97729 with a probability of 0.0000 indicates that the null hypothesis of homoskedasticity was rejected, thus confirming the presence of heteroskedasticity in the model. To correct this violation, Cross-section SUR (Panel Corrected Standard Errors) was applied, to ensure that the estimated standard errors and covariance matrix were robust and that subsequent statistical inferences were valid.

Table 4. Heteroskedasticity likelihood ratio test

Statistics	Value	Type of <i>p</i> -Value	<i>p</i> -Value
<i>F</i> -statistic	11.97729	Prob. <i>F</i> (5,44)	0.0000
Obs* <i>R</i> ²	28.82304	Prob. Chi ² (5)	0.0000
Scaled explained SS	65.14924	Prob. Chi ² (5)	0.0000

Note: Heteroskedasticity test: Breusch-Pagan-Godfrey.

4.2 Test of Hypothesis

The result of the hypothesis testing is shown in Table 5 below.

Table 5. Test of hypothesis

Variable	Coefficient	Std. Error	<i>t</i> -Statistic	Prob.
Cost of sales	8.801653	0.430864	20.42790	0.0000
FSZ	17540014	3583226	4.895035	0.0000
FLEV	8883806	6654906	1.334926	0.1894
FLIQ	-993971.9	314519.5	-3.160287	0.0030
ROA	12436836	7976346	1.559215	0.1268
C	-141,000,000	27731392	-5.093095	0.0000
Effects Specification				
Weighted Statistics				
Statistic	Value	Statistic	Value	
<i>R</i> ²	0.979026	Mean dependent var.	2.765220	
Adjusted <i>R</i> ²	0.974307	<i>S.D.</i> dependent var.	5.586706	
S.E. of regression	0.998348	Sum squared resid.	39.86792	
<i>F</i> -statistic	207.4632	Durbin-Watson stat.	1.721827	
Prob. (<i>F</i> -statistic)	0.000000			

Note: C = Constant term; FSZ = Firm size; FLEV = Firm leverage; FLIQ = Firm liquidity; ROA = Return on assets; Dependent variable: Market value; Method: Panel cross-section SUR (Seemingly Unrelated Regression); Sample: 2015–2024; Periods included: 10; Cross-sections included: 5; Total panel (balanced) observations: 50; Linear estimation after one-step weighting matrix; Cross-section SUR (panel-corrected standard errors) standard errors & covariance (*df.* corrected).

Table 5 presents the panel cross-section SUR regression results, which examined the effect of cost of sales on market value while controlling FSZ, FLEV, FLIQ, and firm profitability. Beginning with the model validity statistics, the adjusted *R*² of 0.974307 indicates that approximately 97.43% of the variations in market value of the listed agricultural firms were explained jointly by cost of sales and the control variables included in the model. This reflects a very high explanatory power, suggesting that the model fit the data well. The Prob. (*F*-statistic) of 0.000000 shows that the overall model was statistically significant at the 5% level, meaning the explanatory variables, taken together, significantly explained changes in firm value. The Durbin-Watson statistic of 1.721827 is close to the benchmark value of 2, indicating the absence of serious autocorrelation in the residuals, thereby supporting the reliability of the estimates.

Still in Table 5, the constant term has a coefficient of -141,000,000 with a probability value of 0.0000. This implied that when cost of sales, FSZ, FLEV, FLIQ, and firm profitability were held constant at zero, market value would decrease by ₦141,000,000. Although a zero value for all explanatory variables might not be economically realistic, the negative and statistically significant constant at the 5% level indicated that other factors not captured in the model exerted a downward baseline effect on market value. The statistical significance confirms that the intercept was meaningfully different from zero.

With respect to cost of sales, Table 5 shows a coefficient of 8.801653 with a probability value of 0.0000. The marginal effect implied that a one-unit increase in cost of sales (₦1,000, since the figures are in thousands) led to an increase of approximately ₦8,801.653 in market value, holding other variables constant. This indicated that increased spending on production and operational activities was associated with a proportionately higher increase

in firm value, suggesting effective cost utilization that translated into revenue growth and investor confidence. Given that the p -value was less than 0.05, the effect was statistically significant at the 5% level. Therefore, the null hypothesis of no significant effect of cost of sales on firm value was rejected, and it was concluded that cost of sales exerted a positive and significant effect on the market value of listed agricultural companies in Nigeria.

Regarding FSZ, the coefficient in Table 5 is 17,540,014 with a probability value of 0.0000. The marginal implication was that a one-unit increase in the natural logarithm of total assets increased market value by ₦17,540,014, *ceteris paribus*. This substantial positive coefficient suggested that larger firms, in terms of asset base, tended to command higher market valuations, possibly due to economies of scale, stronger market presence, and improved access to financing. Since the p -value was below 0.05, the effect of FSZ on market value was positive and statistically significant at the 5% level.

For FLEV, the coefficient is 8,883,806 with a probability value of 0.1894 as shown in Table 5. The marginal effect indicated that a one-unit increase in the leverage ratio increased market value by ₦8,883,806, assuming other factors remained constant. Although the direction of the effect was positive, implying that increased use of debt financing might enhance firm value through tax shields or expansion financing, the p -value exceeded 0.05. This meant the effect was not statistically significant at the 5% level, and thus there was insufficient evidence to conclude that leverage significantly affected market value within the study period.

In the case of FLIQ, Table 5 reports a coefficient of -993,971.9 with a probability value of 0.0030. The marginal implication was that a one-unit increase in the liquidity ratio reduced market value by approximately ₦993,971.9, holding other variables constant. This negative coefficient suggested that excessively high liquidity might reflect idle resources or inefficient asset utilization, which could diminish investor perception of value. Since the p -value was less than 0.05, the negative effect was statistically significant at the 5% level, indicating that liquidity exerted a significant inverse effect on firm value.

Finally, for firm profitability (ROA), the coefficient in Table 5 is 12,436,836 with a probability value of 0.1268. The marginal effect showed that a one-unit increase in ROA increased market value by ₦12,436,836, *ceteris paribus*. Although the positive coefficient suggested that higher profitability tended to enhance firm value, the p -value was greater than 0.05, indicating that the effect was not statistically significant at the 5% level. Therefore, profitability did not exert a statistically significant effect on market value within the context of this study.

4.3 Discussion of Findings

The finding that the cost of sales had a positive and significant effect on firm value indicated that, for Nigerian firms ($\beta = 8.801653$, $p = 0.0000$), the expenses related to producing and delivering goods or services had a direct and beneficial impact on the overall value of the company. Theoretically, cost of sales precedes firm value because production and operating expenditures are incurred in the process of generating revenue, which subsequently determines profitability and ultimately shapes investors' valuation of the firm. From a production and value-creation perspective, efficient allocation of resources to raw materials, labor, and inventory management enhances output and sales performance, thereby transmitting cost decisions into market capitalization rather than the reverse. Furthermore, market value is largely an outcome of investors' expectations about future cash flows, which are fundamentally driven by operational performance indicators such as cost efficiency, hence supporting a unidirectional causal pathway from cost of sales to firm value.

This result is consistent with studies that have shown how managing the cost of sales could lead to increased profitability and, consequently, higher firm value. For example, Giwa & Dibua (2025) found that cost of sales positively impacted shareholder returns in Nigerian companies; therefore they recommended that supply chain managers prioritized efficient sourcing of raw materials and production technologies. King et al. (2025) revealed a significant positive relationship between cost of sales and shareholder returns among listed manufacturing firms, thus reinforcing the idea that managing costs could drive financial success. Similarly, Akinleye & Fakorede (2025) noted that material and overhead costs had a significantly positive impact on profitability; they showed how strategic management of costs could influence a firm's financial outcomes. Additionally, Adesina & Tihamiyu (2025) identified that production costs (if effectively managed) could directly enhance profitability and firm value, especially when incorporating modern cost management techniques like lean manufacturing and AI-driven automation. Therefore, the positive relationship between cost of sales and firm value in this study aligns with the broader empirical findings that effective cost management contributed to better financial performance and shareholder value.

5. Conclusions and Recommendations

It was found that cost of sales had a positive and significant effect on firm value, as indicated in Nigerian firms ($\beta = 8.801653$, $p = 0.0000$). Cost of sales represents direct production and procurement expenses incurred to generate revenue; therefore, its effect on firm value operates through its influence on output, sales growth, and expected future cash flows. When firms increase production to meet higher demand, cost of sales naturally rises,

and if the additional output generates proportionately greater revenue and profit, investors may interpret this as growth potential, leading to higher market valuation. In this sense, a positive effect may reflect scale expansion rather than strict cost control. Larger production volumes can create economies of scale, strengthen competitive positioning, and improve long-term expectations of earnings; all of which enhance firm value. Consequently, the observed positive coefficient might indicate that increased operational activity and revenue-generating capacity could drive valuation, while cost efficiency determined the extent to which rising costs translated into sustainable value creation.

Based on the above, the management of listed agricultural companies was advised to strengthen structured cost management practices that focus on efficient procurement of raw materials, improved inventory control, and optimized production processes, so that spending on cost of sales continues to support revenue growth and translate into higher firm value rather than creating unnecessary operational waste.

5.1 Contribution to Knowledge

This study contributes to the literature by directly addressing gaps left by previous research via examining the effect of cost of sales on the firm value of listed agricultural companies in Nigeria. Unlike earlier studies that focused primarily on firms of manufacturing and consumer goods, this study investigated agricultural companies that had unique operational, market, and production characteristics. While some studies considered cost and competitiveness in agriculture, their focus was on farmers' export decisions rather than firm-level financial performance or market-based value measures. This research integrated both accounting-based and market-based indicators of firm value and controls for FSZ; it employed panel data techniques with corrections for heteroskedasticity and cross-sectional dependence over a ten-year period. By doing so, it provided empirical evidence on how cost of sales, as a key operational resource, drove profitability and firm value in the agricultural sector, thus filling a critical gap in understanding cost management and value creation in listed agricultural companies in Nigeria.

5.2 Limitations of the Study and Suggestions for Further Studies

This study was limited to the financial data of five listed agricultural companies in Nigeria, which might not be able to represent all agricultural firms or other sectors of the economy. The research relied entirely on secondary data from annual reports, so any errors or omissions in these reports could affect the accuracy of the findings. The period of analysis was ten years, which might not fully capture long-term trends or sudden changes in the market. Other factors that could influence firm value, such as government policies, market conditions, or external economic shocks, were not included in the analysis. Additionally, the study focused only on cost of sales and controls on total assets, so the influence of other operational or financial variables on firm value was not examined. These limitations might affect the generalization of the results to other companies or contexts.

Future studies could expand the scope by including more agricultural companies or firms from different sectors to provide a broader understanding of how cost of sales affects firm value. Researchers could also examine additional financial and operational variables, such as debt levels, marketing expenses, or investment in technology, to see how they interact with cost of sales in influencing firm performance. Longitudinal studies covering longer periods may help capture the impact of economic cycles or changes in industry regulations. Primary data, such as surveys or interviews with managers, could provide more detailed information on how decisions of cost control are made in practice. Comparative studies between agricultural firms and other sectors could also help determine if the findings are specific to agriculture or could be applied more widely. This would provide stronger evidence for strategies that enhance firm value.

Author Contributions

Conceptualization, A.K.I., F.O.O., and G.O.N.; methodology, G.O.N.; software, G.O.N.; validation, A.K.I. and F.O.O.; formal analysis, G.O.N.; investigation, G.O.N.; resources, A.K.I. and F.O.O.; data curation, G.O.N.; writing—original draft preparation, G.O.N.; writing—review and editing, A.K.I. and F.O.O.; visualization, G.O.N.; supervision, A.K.I. and F.O.O.; project administration, A.K.I.; funding acquisition, A.K.I. and F.O.O. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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